NLO1 – THE EXACT NONLINEAR RENORMALIZATION STEP

Abstract. This focused outline develops the exact nonlinear renormalization step in the QED3 construction. It states the conditional one-step theorem, its full activity and parameter hypotheses, and the supplied proof tasks for the exact density, two finite localizations, connected logarithm, symmetries and coupling estimates. The general localization and connected-expansion lemmas are separated from their chamber application.

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1. Introduction and target statement

NLO1 is the focused nonlinear-step outline within RGO1, the renormalization stage of QEO1. It draws from Sections 13 and 14 of the main QED3 manuscript, preserving the statement identifiers S12–S15. The target below imports the existing block and normalization estimates and specifies the remaining one-step proof.

Broader plans: RGO1/S15 and QEO1/S15. The overview boxes and statement-page return links are clickable.

S15. The symmetric correction and the exact nonlinear step

Model-specific. Overview Target.

Proof dependencies. P, RNO1, S07, M20, S12, S13, and S14.

Standing parameters and imported hypotheses. Fix $0 < \rho < 1$, odd $L \geq 4^8$, $M = L^m \geq L^2$, $M > 4L$, and $k \geq 0$. Put $\eta = L^{-k}$, $e = e_k \in (0, 1)$, $e_{k+1} = \sqrt{L} e_k$, and $0 < \mu \leq 1/L$; the scaled reference mass is $L\mu$. Both charges are below the common ceiling of the imported primitive estimates in P. Require $\ell/(LM)$ and $\ell/(L^2 M)$ to be odd integers at least 13. Fix an integer $r \geq 2$. For $v = -\log e$, put

$$r_k = \min\{L^i : i \geq 0 \text{ an integer},\ L^i \geq v^r\}, \qquad \ell \geq M r_k.$$

This implies the determinant volume condition $\ell \geq M v^2$; no stopping rule or tuned trajectory is assumed.

Use the complete activity algebra, coefficient norm, local gauge domains and generator actions of D. Barred and unbarred generators are independent; all spinors are antiperiodic in each coordinate.

The quantitative constants are

$$\bar{\varepsilon}=1/200,\qquad 0<\varepsilon\le\bar{\varepsilon},\qquad 1-\varepsilon<\alpha<1,$$
$$c_X=54,\quad \kappa_0=54\log 72,\quad K_0=\exp(\kappa_0)/36,\quad c_{\mathrm{rb}}=c_XK_0,\quad c_{\mathrm{fl}}=61c_{\mathrm{rb}},$$
$$c_C=3L^{9/2},\quad b=1,\quad b_L=b/L,\quad b_-=1-L^{-1},$$
$$C_T=2\exp(3/2)[1+b_-^{-1}(54L\exp(6)+12\exp(1)+1)],$$
$$c_S=10^5c_X^5K_0^5\exp(6\kappa),\qquad \kappa_w=\kappa+\kappa_0+3.$$

The primitive constants $C_w,\gamma_w,\alpha_0>0$, the photon local-gauge constant C_B, and the two exponential-sum bounds c_γ are those specified in P; $C_w,c_\gamma\ge1$. Assume the photon and fermion conclusions of P for one system with these common constants, the weighted algebra of S07, the geometry of M20, and the extraction, reconstruction and scaling conclusions of RNO1.

The normalization and scaling assumptions are retained in full:

$$L^{1/8-3\bar{\varepsilon}/2}\ge2400,\qquad e^{\varepsilon}\le\tfrac14,\qquad 160L^{1+3\varepsilon/2}Me^{\varepsilon}\le1,$$
$$9ML^{\varepsilon}e^{2\varepsilon}\le1,\qquad 90c_XLMe^{\varepsilon}\le1,$$
$$\kappa\ge7\kappa_0+7,\qquad L(\kappa-6\kappa_0-4)-1\ge\kappa.$$

For the two localizations require

$$C_w\ge4\exp(2\gamma_w)M^{27\alpha_0},\qquad M^{\alpha_0}-1\ge\exp\left((2L+1)^3\kappa+\log(6c_X)\right),$$
$$c_{\mathrm{fl}}e^{\varepsilon}\le1,\quad 3Me^{\varepsilon}\le1,\quad 2c_CC_w^2c_\gamma^2e^{\varepsilon/4}\le1,$$
$$16c_Xc_\gamma C_wC_T^2M^3e^{\varepsilon/2}\le1,\quad 96M^3e^{1/4}\le1,\quad 32C_wC_Te^{3\varepsilon/2}\le1,$$
$$88M^3e^{1/4-6\varepsilon}\le1,\qquad 2e^{5\varepsilon/2}\le1,\qquad 112C_w^2c_\gamma e^{2\varepsilon}\le3.$$

The connected expansion, scaling and determinant reblocking also use

$$c_Se^{1/4-5\varepsilon}\le1,\qquad c_SL^3e^{\varepsilon}\le1,\qquad 8c_{\mathrm{rb}}L^3C_dM^3e^{2\varepsilon}\le1,$$

where C_d is the determinant constant of P. For fixed positive exponents p_0,p, put $p_{0,k}=v^{p_0}$, $p_k=v^p$. The photon box and domain bounds and the determinant gain require

$$p_{0,k}\le e^{-\varepsilon/4},\qquad 12\exp(-p_{0,k}^2/2)\le e^7,\qquad 8C_Bp_k\le e^{-3/4+3\varepsilon},$$
$$8e^{3/4-7\varepsilon/2}\le1,\qquad L^{3/4}\ge8/3,\qquad v\ge9,\qquad 8M(v/4+4)e^{\varepsilon}\le1,$$
$$e^{1/4}+e^{\varepsilon}<1,\qquad 4e^{1/4+2\varepsilon}+2e^{3\varepsilon}\le1.$$

These are simultaneous hypotheses on the one step, with the primitive admission conditions already included in P. They retain the three normalization condition sets in QEM1, `eq:rg-adjustment-charge-condition`, `eq:rg-contraction-parameters`, and `eq:rg-contraction-polymer-parameters`; the photon and connection domain conditions in Sections 10 and 12; and `eq:determinant-gain-conditions` and `eq:rgnl-fixed-smallness`.

Data, fields and the two radii. The ordinary and pair weights are $h_k=(e_k^{-1/4},e_k^{-1/4+\varepsilon})$. Set

$$w_k^m=e_k^{3/4-8\varepsilon},\qquad w_k^E=e_k^{1/4-7\varepsilon},\qquad \mathrm{Ball}_k^x(c)=\{(m,E):|m|\le cw_k^m,\ \|E\|_k\le cw_k^E\}.$$

The data are complex for the nonlinear correction; for the exact real density statement take real m,ε_v and $E\in\mathcal{A}_k^x$. At $k=0$, set $E=0$ and $q=e_{k=0}$. The input and output densities and the normalized block operator $\mathbf{T}_k^x$ are defined explicitly in S12. Their external-only scalar families $P_k(X,C)$ are retained.

The construction to be proved has three inputs, developed in the following statement pages: the exact finite density of S12, the finite activity F and its locality and norm bound in S13, and the

general connected logarithm in S14. For its application set

$$V = F, \qquad H = 36c_X^2 K_0^2 e_k^{1/4-5\varepsilon}, \qquad w = h_k/4, \qquad w_W = \begin{cases} 1, & k \geq 1, \\ \sqrt{48}, & k = 0. \end{cases}$$

The finite bounds and the displayed c_S condition verify the three smallness inequalities in S14. The two finite sum identities identify this logarithm with the density fluctuation factor; this identification is part of the proof below.

Claim. The logarithm has the lattice symmetry with LM-translations, independent local real-gauge covariance, charge conjugation and, for real data, reality. Each monomial is balanced. Define

$$E_k^*(m, E)(Y; A, C) = \sum_{\overline{X}^L = LY} S_L E_k^{\#}(m, E)(X; A_L, C_L).$$

It is a next-level activity and $\|E_k^*(m, E)\|_{k+1} \leq e_k^{1/4-6\varepsilon}$ on the closed data ball of radius 4. It is norm-continuous on each affine data line's closed allowed set and norm-analytic in its interior. On the unit ball,

$$\|E_k^*(m_1, E_1) - E_k^*(m_2, E_2)\|_{k+1} \leq e_k^{1/4-6\varepsilon} \left(\frac{|m_1 - m_2|}{w_k^m} + \frac{\|E_1 - E_2\|_k}{w_k^E} \right).$$

At $k = 0$, omit the activity-difference term. Let E_k^d be the even local determinant primitive and put

$$\delta E_k^{\det}(X; A, C) = E_k^d(X, A + C) - E_k^d(X, C), \qquad E_k^{\det}(X, C) = E_k^d(X, C) - E_k^d(X, 0).$$

The exact output maps are

$$\begin{aligned} E^+ &= \mathcal{L}(RE)^{\mathrm{nat}} + \mathcal{L}\delta E_k^{\det} + E_k^*(m, E), \\ m^+ &= L(m + m(E)), \\ \varepsilon_v^+ &= L^3(\varepsilon_v + \varepsilon_v(E) + \delta\varepsilon_k^0), \\ P_{k+1}(Y, C') &= \sum_{\overline{X}^L = LY} [P_k(X, C'_L) + E_k^{\det}(X, C'_L)]. \end{aligned}$$

For real data in $\mathrm{Ball}_k^x(1)$, every $A' \in \mathrm{Dom}_{k+1}$ with arbitrary harmonic part, and every $C' \in \mathfrak{X}_{k+1}$, they give

$$\mathbf{T}_k^x \mathscr{F}_k^x[\varepsilon_v, m, E] = \mathscr{F}_{k+1}^x[\varepsilon_v^+, m^+, E^+],$$

with scalar families P_k, P_{k+1} on the respective sides. Writing $y_j^x = \int \mathrm{D}\Psi\, \mathscr{F}_j^x$, the exact marginal is

$$y_{k+1}^x(A'; C') = \int_{\mathrm{Box}_k} d\mu_I(\zeta)\, y_k^x(A_k^{\min}(A') + \mathrm{Sq}_k\zeta; C'_L).$$

This uses the unnormalized Gaussian box restriction. The output satisfies $E^+ \in \mathcal{A}_{k+1}^x$ and

$$\begin{aligned} \|E^+\|_{k+1} &\leq c_{\mathrm{fl}} L^{-1/4+2\varepsilon} \|E\|_k + e_{k+1}^{1/4-2\varepsilon} + e_k^{1/4-6\varepsilon}, \\ |m^+ - Lm| &\leq L c_{\mathrm{fl}} e_k^{1/2} \|E\|_k, \\ |\varepsilon_v^+ - L^3\varepsilon_v| &\leq L^3(c_{\mathrm{fl}}\|E\|_k + e_k^7). \end{aligned}$$

At zero $m^+ = Lm$ and the vacuum increment is precisely $L^3\delta\varepsilon_0^0 \in [0, L^3 q^7]$. All operations commute with restriction of the external slot to zero.

Proof work. Pass lattice, gauge and reality actions through finite operations. For charge conjugation the asymmetric unbarred/barred fluctuation substitution is not directly covariant. At indicator parameters 1_U, compare its Wick covariance

$$G_U(B) = H_k(1_U, B)\Gamma_k(1_U, B)\bar{H}_k(1_U, B)^T, \qquad G_U(-B)^T = -\mathsf{C}^{-1}G_U(B)\mathsf{C}^{-1}.$$

Derive equality of all transformed pair contractions and apply Wick; subset inversion then gives covariance of each integrated Mayer activity and hence of the local logarithm. Constant-gauge invariance forces balance. Use the connected bound and reblocking for E_k^*. Fixed-volume polynomial approximants prove norm analyticity. Vary mass and activity separately and apply Cauchy's formula at radii $3w_k^m/|m_1 - m_2|$ and $3w_k^E/\|E_1 - E_2\|_k$, respectively, omitting a variation when its difference is zero. This gives the Lipschitz bound. Evaluate the contour expressions $G_1, \dots, G_5$ defined in S13. Its two exact localization sum identities identify their evaluated sum with the density fluctuation exponent. Use the three evaluation homomorphisms of QEM1 `thm:rgnl-current-step`: evaluation at the outgoing mean, at its shifted fine field, and at the auxiliary coarse field. They agree on the required ordinary and pair generators; evaluation fixes the fluctuation generators. Reconstruction uses $T_k H_k = 1$. Split the determinant quotient, retain adjustment's evaluated equality, and assign every fine polymer once under reblocking. The determinant term needs the full-norm consequence

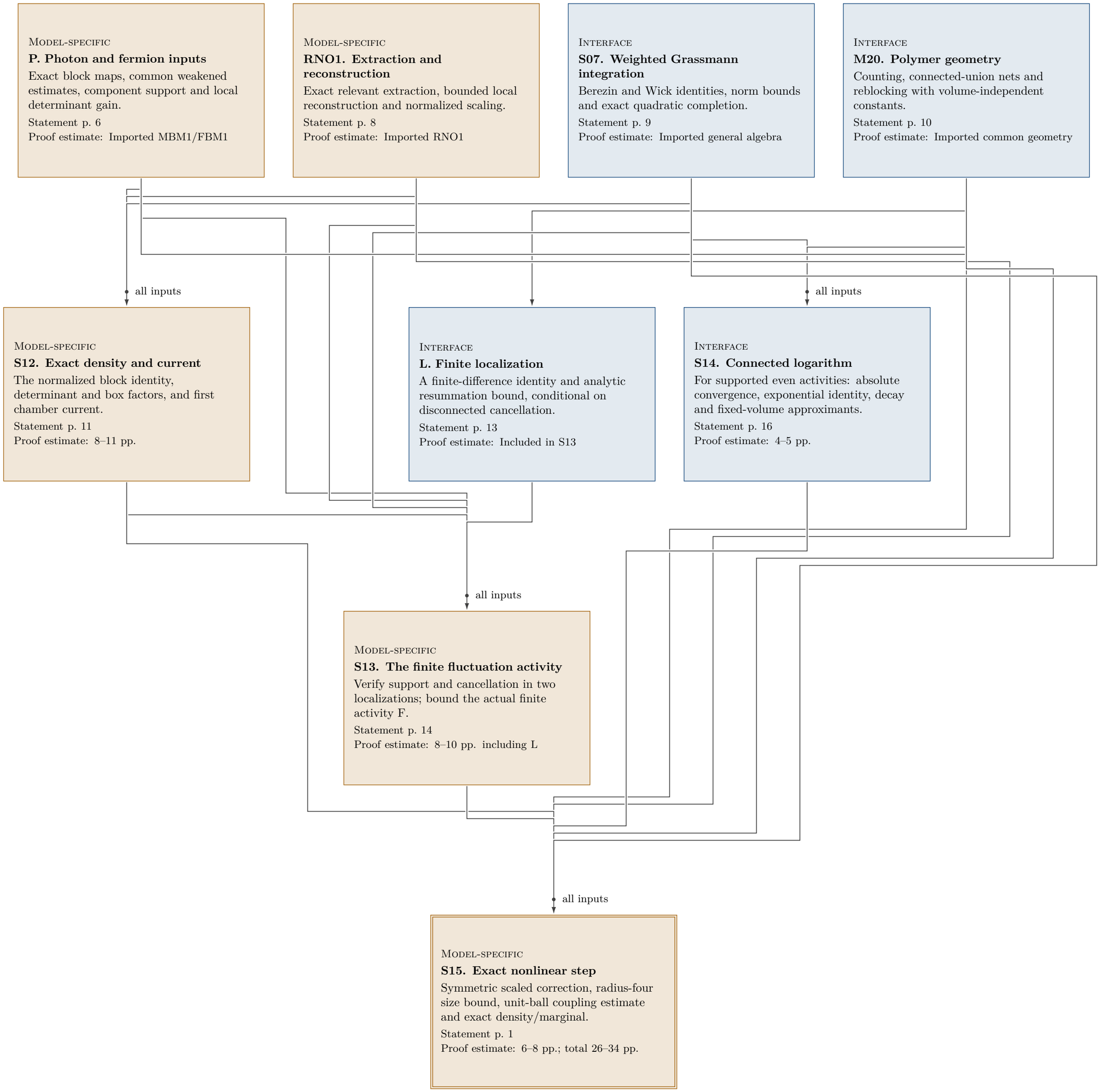
$$\|\mathcal{L}\delta E_k^{\det}\|_{k+1} \le 8c_{\mathrm{rb}}L^3 C_d M^3 e_k^{1/4} \le e_{k+1}^{1/4-2\varepsilon}.$$

Combine this with normalized linear contraction and the nonlinear correction bound. The two forcing terms are retained; the result does not assert a contraction of the entire nonlinear map.
Inputs. Finite localization; connected logarithm; normalized contraction; exact density/current; determinant gain; weighted moments. This gives one exact nonlinear step with size and coupling-difference bounds. Equality across covers and common tuning are subsequent tasks.
Source. PDF pp. 153–159, §§14.5–14.7, Propositions 14.6–14.7, Theorem 14.8, (14.35)–(14.52). TeX 14108–14829, especially `prop:rgnl-scaled-correction` and `thm:rgnl-current-step`. The eventual full proof budget is 26–34 pages for S12–S15 together, with the imported block and normalization proofs counted separately. This document is the focused proof outline.

NLO1 – The exact nonlinear renormalization step

Definitions/setup | Model-specific | Interface (general)

Standing hypotheses and the complete target are on S15; D gives the activity space. P and RNO1 are imported results. S14 is a general theorem: S13 verifies its application at the final all-inputs join.



Arrows record mathematical dependencies; no distinction between proof and application inputs is made here. Every incoming branch at a junction is required together with the standing hypotheses on the statement page.

Click a box for its statement. Return links are navigation, not proof dependencies.

The first-step current from S12 is used in S13. There is no backward dependence from the logarithm to finite locality. Page budgets count the four principal tasks once; imported proofs are separate.

2. P. Imported photon and fermion estimates

Model-specific. Overview Target.

The primitive system used by the step. Assume one system of the following maps on the stated torus and its next scale. This is the joint input from MBO1/MBT and FBO1/FT, under their full hypotheses. The corresponding proof manuscripts are MBM1, Theorems A–B, and FBM1, Theorems A–B and Proposition 6.2. Their estimates supply the stated operators; this outline assumes their common constants and the covariance and component properties used below.

Photon split, box law and domains. The real photon maps satisfy

$$H_k^{\mathrm{ph}} A_k^{\min}(A') = (H_{k+1}^{\mathrm{ph}} A')_L - d\omega_k, \qquad \mathrm{Sq}_k = C_{\mathrm{ph}} C_k^{1/2},$$

where ω_k is real and periodic. Here C_{ph} denotes the source's coordinate map C, distinguished from the external connection C in the activity notation. For real unit bond fields define

$$\mathrm{Dom}_j = \{A : H_j^{\mathrm{ph}} A \in \tfrac{1}{4}\widetilde{R}_j\}.$$

The scaled field and external domain satisfy

$$(H_{k+1}^{\mathrm{ph}} A')_L \in \tfrac{1}{4} L^{-3/4} \widetilde{R}_k \subset \widetilde{R}_k^-, \qquad (\mathfrak{X}_{k+1})_L \subset \mathfrak{X}_k.$$

Every $A' \in \mathrm{Dom}_{k+1}$, with arbitrary harmonic part, and every box coordinate ζ give $A_k^{\min}(A') + \mathrm{Sq}_k \zeta \in \mathrm{Dom}_k$. The box is $\mathrm{Box}_k = \{|\zeta(\Upsilon)| \le p_{0,k}\}$. Its normalized law is the product probability law

$$d\widehat{\mu}_k = \frac{\mathbf{1}_{\mathrm{Box}_k}}{\mu_I(\mathrm{Box}_k)} d\mu_I, \qquad \mu_I(\mathrm{Box}_k) = \exp(-n_k \delta\varepsilon_k^0), \quad n_k = \ell^3, \quad 0 \le \delta\varepsilon_k^0 \le e_k^7.$$

The coordinate row bound is $\sup_b \sum_\Upsilon |C_{\mathrm{ph}}(b,\Upsilon)| \le c_C$. The constant C_B bounds the three norms of a local real representative of $H_k^{\mathrm{ph}} A$ by $C_B \|dA\|_\infty$; it depends only on the fixed L, M, α.

Analyticity, decay and component restrictions. The photon reconstruction and square root have independent parameter cubes of half-side M and LM; the fermion parameters are attached to partition M-cubes. All parameters range over $|s_{\mathrm{pc}}| \le M^{\alpha_0}$. The photon reconstruction, its first derivative and its short Hölder derivative, and the square-root kernel have bounds $C_w \exp(-\gamma_w d)$ with their stated index measures. Choose $c_\gamma \ge 1$ so that

$$\sup_b \sum_{b'} e^{-\gamma_w d(b,b')} \le c_\gamma, \qquad \sup_\Upsilon \sum_{\Upsilon'} e^{-\gamma_w d(\Upsilon,\Upsilon')} \le c_\gamma.$$

The fermion $H_k, \bar{H}_k$, their short covariant quotients, and Γ_k have the same exponential bound with constant C_w. Their local Dirac kernels satisfy

$$|M_k(X,B)(z,z')| \le C_w \exp[-\gamma_w d(z,z') - \kappa_w d_M(X)].$$

All these families are analytic in the allowed connections and in the interior parameter polydisc, and continuous on the parameter closure. They read only the stated local bonds and coordinates. Setting the parameters outside a cube union to zero restricts them to its active components. An H_k row at (c,g) vanishes unless every cube meeting $\{|c'-c|_\infty \le 2L\}$ is active. The covariance preserves components; its parameter-independent diagonal part is not required to have this stronger vanishing property.

At parameter one the families are the exact operators, and

$$\sum_X M_k(X,B) = b_k - D_k(B), \qquad H_k = b_k S_\infty \mathcal{Q}_k^\sharp, \quad \mathcal{Q}_k = Q_k J^*, \quad D_k = b_k - b_k^2 \mathcal{Q}_k S_\infty \mathcal{Q}_k^\sharp \quad (k \ge 1).$$

Here $S_\infty = (D_B + \mu + b_k \mathcal{Q}_k^\sharp \mathcal{Q}_k)^{-1}$, $b_k = (1-L^{-1})/(1-L^{-k})$, and the nested averages and algebraic reverse phases are those of FBO1/F0. The barred map is $\bar{H}_k(B) = \mathsf{C}^{-1} H_k(-B)\mathsf{C}$. At zero, retain the diagonal photon families, $H_0(1,B) = 1$, the supported weakened H_0, and the full chamber covariance Γ_0; there is no M_0. All fine kernels use column measure $\eta^3/48$.

The families have their actual lattice, real-gauge, transpose and reality covariance, with parameters transformed as well. These are the identities in QEM1 Sections 10.4 and 11.1–11.3, including the exchange of left and right propagators under transposition. The identity Wick measure and photon box law have the matching covariance. Charge conjugation of the integrated nonlinear activity still requires the Wick comparison in S15.

Local determinant input. Assume the analytic even local primitive $E_k^d(X,B)$ with real-gauge invariance, lattice covariance and analytic reality, and

$$\frac{\delta Z_k(B)}{\delta Z_k(0)} = \exp \sum_X [E_k^d(X,B) - E_k^d(X,0)], \qquad |E_k^d(X,B)| \le C_d M^3 e^{-(\kappa+1)d_M(X)}.$$

On the two-slot domains and under the determinant conditions in S15, its difference obeys

$$|E_k^d(X,A+C) - E_k^d(X,C)| \le 8 C_d M^3 e_k^{1/4} e^{-\kappa d_M(X)}.$$

This pointwise bound is the input to the reblocked full-norm estimate in S15.

Source and use. QEM1 Sections 10–11, especially `prop:photon-domain-preservation`, `prop:photon-weakened-bounds`, `prop:weakened-polymer-propagators`, `prop:coarse-local-family-bounds`, `prop:local-determinant-expansion` and `prop:determinant-small-field-gain`; the joint hypotheses are restated at TeX 13098–13128. Exact density uses the photon split and full-strength fermion maps; finite localization uses the weakened support properties; the symmetry proof uses their covariance. Equality of independently constructed systems on different covers is a later theorem.

3. RNO1. Extraction, reconstruction and normalized scaling

Model-specific. Overview Target.

Imported conclusions. Use RNO1/T under the complete parameter conditions displayed in S15, on the activity space of D. The linear extraction functionals obey

$$|\varepsilon_v(E)| \le K_0 M^{-3}\|E\|_k, \qquad |m(E)| \le K_0 M^{-3} e_k^{1/2}\|E\|_k.$$

Writing $E^r = RE$, subtraction of the vacuum and mass terms is an exact evaluated identity. The shortest-chain adjustment $(RE)^{\mathrm{nat}}$ preserves that evaluation and makes both the vacuum and the complete ordinary quadratic kernel vanish on small polymers at zero connection.

For positive levels the local reconstruction T_k satisfies $T_k(B)H_k(B) = 1$ on evaluated fields. Its ordinary interior rows and external-neighbour boundary pair rows have coefficient sums bounded by C_T. Giving auxiliary fields weight $C_T h_{k,1}$ makes this substitution contractive in the weighted coefficient norm. It creates only type-3 boundary pair legs. The construction and its covariance are RNO1/S10; no reconstruction is used at level zero.

Scale connections and generators by

$$B_L(b) = L^{-1/2}B(b/L), \qquad \psi_L = L^{-1}\psi(\cdot/L), \qquad \chi_L = L^{-1-\alpha}\chi(\cdot/L, \cdot/L).$$

Let S_L be the resulting algebra map and define

$$(\mathcal{L}E)(Y; A', C') = \sum_{\overline{X}^L = LY} S_L E(X; A'_L, C'_L).$$

The scaled-domain inclusion gives $(\widetilde{R}_{k+1})_L \subset \widetilde{R}_k^-$ and $(\mathfrak{X}_{k+1})_L \subset \mathfrak{X}_k$. The normalized linear map satisfies

$$\|\mathcal{L}(RE)^{\mathrm{nat}}\|_{k+1} \le c_{\mathrm{fl}} L^{-1/4+3\varepsilon/2}\|E\|_k \le c_{\mathrm{fl}} L^{-1/4+2\varepsilon}\|E\|_k.$$

These maps preserve the required support, symmetry and reality conditions and commute with restriction $C = 0$.

Use and source. Extraction enters S12; bounded reconstruction enters the first finite localization S13; adjustment, scaled domains and normalized contraction enter S15. The nonlinear correction is constructed after these imports. QEM1 Section 12 and Section 13.4; RNO1/S09, RNO1/S10 and RNO1/S11. Their proofs are a separate assignment; their page budgets are not added to the nonlinear proof budget.

4. S07. Weighted Berezin integration and exact block completion

Interface. Overview Target.

Objects and hypotheses. Let I be a nonempty finite ordered mode set with positive weights $\nu(t)$, and independent generators $W(t), \bar{W}(t)$. An operator kernel acts using the column weight $\nu(t')$; its quadratic form is $\sum_{t,t'} \nu(t)\nu(t')\bar{W}(t)O(t,t')W(t')$. Define $\mathrm{D}W$ as extraction of the coefficient of $\prod_t W(t)\bar{W}(t)$, multiplied by $\prod_t \nu(t)^{-1}$. For invertible C, put

$$\mathrm{d}\mu_C(W) = (\det C)\,\mathrm{D}W \exp[-\langle \bar{W}, C^{-1}W\rangle_\nu].$$

For any finite families of Grassmann generators give an ordered monomial the product of its generator weights, and a polynomial the sum of the moduli of its coefficients times those weights. This is equivalently the absolute antisymmetric-kernel norm with the index measures and the factors $1/n!$.
Claim. The norm is submultiplicative. With $w_W = \max_t \nu(t)^{-1/2}$,

$$\left\| \int \mathrm{d}\mu_I(W) F \right\|_w \le \|F\|_{w,w_W},$$

$$\int \mathrm{d}\mu_C(W) F(\bar{W}, W) = \int \mathrm{d}\mu_I(W) F(\bar{W}, CW),$$

$$\int \mathrm{d}\mu_C(W) W(t)\bar{W}(t') = C(t,t'), \qquad \int \mathrm{D}W\, e^{-\langle \bar{W}, OW\rangle_\nu} = \det O.$$

The last formula does not require O invertible. Disjoint mode sets factor, with external generators retained as coefficients. For linear odd fields $v_1, \dots, v_{2r}$, prove the Pfaffian Wick formula; odd moments vanish. If a linear substitution sends family φ_i to $\sum_j O_{ij}\theta_j$, its norm is at most one whenever

$$\sum_j \left(\sup_t \sum_s |O_{ij}(t,s)| \nu'_j(s) \right) w'_j \le w_i$$

for each occurring input row.

For two finite spinor spaces with their specified pairings, let $K = D + b_L A^\sharp A$ be invertible, $\Gamma = K^{-1}$, and set

$$\Psi^c = b_L \Gamma A^\sharp \Phi, \quad \bar{\Psi}^c = b_L \Gamma^T A^T \bar{\Phi}, \quad D' = b_L - b_L^2 A \Gamma A^\sharp.$$

Put $s = b_L A^\sharp \Phi$, $\bar{s} = b_L A^T \bar{\Phi}$. Then completion of the block quadratic form gives

$$\int \mathrm{D}\Psi \, \exp\{-\langle \bar{\Psi}, K\Psi\rangle + \langle \bar{\Psi}, s\rangle + \langle \bar{s}, \Psi\rangle - b_L\langle \bar{\Phi}, \Phi\rangle\} F(\bar{\Psi}, \Psi)$$

$$= \det K\, e^{-\langle \bar{\Phi}, D'\Phi\rangle} \int \mathrm{d}\mu_I(W) F(\bar{\Psi}^c + \bar{W}, \Psi^c + \Gamma W).$$

Here $A^\sharp$ is the specified algebraic reverse map. In the chamber application it is the transpose of the averaging map at the reversed connection. No Hermitian conjugation of Grassmann fields is used.
Proof work. Compute each paired monomial in the identity Gaussian. Its modulus is exactly covered by w_W. A linear change of the unbarred variables multiplies the top coefficient by the operator determinant. Left Grassmann differentiation and integration by parts fix the Wick signs. Bound images of generators for the substitution estimate. Translate by the displayed means, keeping odd factors in their original order, and cancel both cross terms.
Inputs. Finite exterior algebra and determinants only. Exports: weighted moments, substitutions, exact Gaussian block identity. In the application $\nu = 1/48$ at level zero, so $w_W = \sqrt{48}$ and the identity two-point kernel is $48\delta_{tt'}$. At positive levels $\nu = 1$.
Source. PDF pp. 107–110, §§11.4–11.5, Proposition 11.4, (11.20)–(11.29). TeX 9514–9754.

5. M20. Polymer counting, connected unions and reblocking

Interface. Overview Target.

Objects and hypotheses. Let $L \geq 5$ be odd, $M \geq L^2$ a power of L, and let a cubic torus side ℓ have $\ell/(LM)$ an odd integer at least 13. A polymer X is a nonempty face-connected union of M-cubes. Define $Md_M(X)$ as the infimum length of a connected net meeting their closed cubes. A net is a union of images of specified straight segments; length is the sum of their represented Euclidean lengths, with the infimum also over those representations. Let $\overline{X}^{\,L}$ be its closure in LM-cubes.
Claim. With $c_X = 54$, $\kappa_0 = 54 \log 72$, $K_0 = e^{\kappa_0}/36$, prove, with $|X|_M$ the number of cubes,

$$d_M(X) \leq |X|_M - 1, \quad |X|_M \leq c_X(d_M(X)+1), \quad d(p,q) \leq M(d_M(X)+2) \quad (p,q \in X),$$
$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \leq K_0, \qquad \sum_{X \subset Y} e^{-\kappa_0 d_M(X)} \leq K_0 |Y|_M,$$

where sums run over nonempty connected polymers and Y is any cube union. For reblocking require also that ℓ/L has the same side conditions, and prove

$$Ld_M(Y) \leq d_M(X) \quad (\overline{X}^{\,L} = LY), \qquad \sum_{\overline{X}^{\,L} = LY} e^{-\kappa' d_M(X)} \leq c_X K_0 L^3 e^{-L(\kappa' - \kappa_0 - 1) d_M(Y)}.$$

Here $\kappa' \geq \kappa_0$.
Proof work. Construct a rooted cube tree from a near-minimal represented net, count its finite neighbours and tree codes, and integrate the exponential tail. Scale the represented segments before taking the infimum for reblocking. Joining represented nets along a tree of incompatible polymers adds less than three units of rescaled length per edge. All constants must remain independent of torus side.
Inputs. Cubic geometry only. This is an upstream geometric lemma used by localization, cover comparison and polymer logarithms; it has no field-theoretic input.
Source. Manuscript pp. 261–263, Appendix A and Proposition A.1.

Supporting proof: 2–3 pages if not imported; source Appendix A is reused.

6. S12. The exact density identity and the first chamber current

Model-specific. Overview Target.

Proof dependencies. P, S07, and extraction in RNO1.

Objects and hypotheses. Incoming unit spinors have four components and counting measure at $k \geq 1$; at $k = 0$ there are 192 components per photon site and measure $1/48$. The outgoing L-spaced space has four components and measure L^3. Put $\mathcal{A}_k = Q(A)$ at positive levels and $\mathcal{A}_0 = Q(A)J^*$ at zero, with algebraic reverse $\mathcal{A}_k^\sharp = \mathcal{A}_k(-A)^T$. Let $\mathcal{D}_k = D_k$ for $k \geq 1$, $\mathcal{D}_0 = D_A + \mu$, and

$$K_k = \mathcal{D}_k + b_L \mathcal{A}_k^\sharp \mathcal{A}_k, \quad \Gamma_k = K_k^{-1}, \quad \delta Z_k = \det K_k.$$

Fix odd $L \geq 5$, $M = L^m \geq L^2$, and $\ell/(LM)$ odd and at least 13, with the outgoing side also admissible. The normalized density has exponent

$$-\langle \bar{\Psi}, \mathcal{D}_k(A + C)\Psi \rangle + I_k[m, E](A, C; \Psi) - \varepsilon_v n_k + \sum_X P_k(X, C), \qquad n_k = \ell^3,$$

multiplied by $\det \mathcal{D}_k(0)^{-1}$. Here I_k evaluates $-m\langle \bar{\psi}, \psi \rangle + \sum E(X)$ at $\psi = H_k(A+C)\Psi$ and its barred and covariant pair fields. The density's first argument is a real coarse photon field reconstructed by H_k^{ph}. Use the activity class $\mathcal{A}_k^{x,\mathbb{C}}$ defined by the complete domains and generator covariance equations in D; the scalar data m, ε_v may be complex in this exact identity. Assume the exact photon split $H_k^{\mathrm{ph}} A_k^{\min}(A') = (H_{k+1}^{\mathrm{ph}} A')_L - d\omega_k$, $\mathrm{Sq}_k = C_{\mathrm{ph}} C_k^{1/2}$, its box law, all domain inclusions, and the fine/coarse resolvent and field-map definitions. In particular $H_{k+1}^{\mathrm{ph}} A'$ has one-quarter local real-gauge bounds with thresholds $e_{k+1}^{-3/4+i\varepsilon}$, while C' has thresholds $e_{k+1}^{-1+2\varepsilon}, e_{k+1}^{-1+3\varepsilon}$ on its sup and first derivative. The admitted box backgrounds $B, B + Z, B + Z - d\omega_k$ and their negatives lie in the proved open inverse domain. Use $0 < \mu \leq 1/L$, $e_{k+1} = \sqrt{L} e_k$, and zero input activity at $k = 0$.

Density and block operator. Write the displayed normalized density as $\mathscr{F}_k^x[\varepsilon_v, m, E](A_k, C; \Psi)$, where the fine first connection in its exponent is $H_k^{\mathrm{ph}} A_k$. The mass pairing in I_k is always the chamber pairing $\eta^3 \, 48^{-1} \sum_{c,g,\beta} \bar{\psi}_\beta \psi_\beta$. Let

$$A_k(\zeta) = A_k^{\min}(A') + \mathrm{Sq}_k \zeta, \quad \mathscr{B}_k = H_k^{\mathrm{ph}} A_k^{\min}(A'), \quad C = C'_L,$$
$$\mathcal{Q}_{\mathrm{bl}} = -b_L \langle e^{-ie_k\omega_k} \bar{\Psi}'_L - \mathcal{A}_k(-\mathscr{B}_k - C)\bar{\Psi}, \, e^{ie_k\omega_k} \Psi'_L - \mathcal{A}_k(\mathscr{B}_k + C)\Psi \rangle_L.$$

The transform in the target statement is

$$(\mathbf{T}_k^x F)(A', C'; \Psi') = b^{-4n_{k+1}} \int_{\mathrm{Box}_k} d\mu_I(\zeta) \int \mathrm{D}\Psi \, e^{\mathcal{Q}_{\mathrm{bl}}} F(A_k(\zeta), C'_L; \Psi).$$

The outgoing fields scale by L^{-1}, as in RNO1.

Claim. For real B and $\mu > 0$, prove invertibility of $D_B + \mu$ and $K_k(B)$ at all levels, and of $D_k(B)$ for $k \geq 1$. At positive levels,

$$D_k(B)^{-1} = b_k^{-1} + \mathcal{Q}_k(B)(D_B + \mu)^{-1} \mathcal{Q}_k^\sharp(B), \qquad \mathrm{Re}\, D_k^{-1} \geq b_k^{-1}.$$

Schur elimination and scaling identify the outgoing quadratic density, the induced mean field and

$$\det D_{k+1}(B') = b^{4n_{k+1}} \frac{\det \mathcal{D}_k(B'_L)}{\delta Z_k(B'_L)}.$$

Let $A = (H_{k+1}^{\mathrm{ph}} A')_L$, $C = C'_L$, $B = A + C$, $Z = H_k^{\mathrm{ph}} \mathrm{Sq}_k \zeta$, $m' = m + m(E)$, $E^r = RE$. With the block means $\Psi^c, \bar\Psi^c$, put $\Psi^f = \Psi^c + \Gamma_k(B) W$, $\bar\Psi^f = \bar\Psi^c + \bar W$. The transformed density is exactly

$$\begin{aligned}\det D_{k+1}(0)^{-1} \exp\{ &- \langle \bar\Psi', D_{k+1}(B')\Psi'\rangle - Lm'\langle \bar\psi', \psi'\rangle \\ &- L^3(\varepsilon_v + \varepsilon_v(E) + \delta\varepsilon_k^0) n_{k+1} + \sum_X E^r(X; A, C; \psi^{0,\#}) + \sum_X P_k(X, C)\} \\ &\cdot \frac{\delta Z_k(B)}{\delta Z_k(0)} \, \Xi_k,\end{aligned}$$

where $B' = H_{k+1}^{\mathrm{ph}} A' + C'$, $\psi^{0,\#}$ denotes the scaled outgoing fields, and

$$\begin{aligned}\Xi_k &= \int \mathrm{d}\mu_I(W) \, \mathrm{d}\widehat{\mu}_k(\zeta) \, e^{V_k}, \\ V_k &= \mathrm{Act}[m', E^r](A + Z, C; \Psi^f) - \mathrm{Act}[m', E^r](A, C; \Psi^f) \\ &\quad + I[m', E^r](A, C; \Psi^f) - I[m', E^r](A, C; \Psi^c).\end{aligned}$$

Here Act includes the kinetic quadratic term. The photon law $\widehat{\mu}_k$ is normalized on the coordinate box; the box probability is exactly $\exp(-n_k \delta\varepsilon_k^0)$. The block transform has normalization $b^{-4n_{k+1}}$; its outgoing marginal is the unnormalized box integral of the incoming marginal.

First-step model verification. At $k = 0$, $H_0 = 1$ and $q = e_{k=0}$. The external stencil matrix is $M_3(c, g) = 3(\rho - \gamma \cdot g e_1)$, and $c_3 = c + g e_1$ at this mesh. On a cube define

$$\begin{aligned}V_0(Z, \square) = \tfrac{1}{2} \int_{\square} &[\bar\psi(x)^T M_3(x)(e^{iqZ(c,c_3)} - 1)(\psi(x) + \chi(x, x^{(3)})) \\ &+ (\bar\psi(x) + \bar\chi(x, x^{(3)}))^T M_3(x^{(3)})(e^{-iqZ(c,c_3)} - 1)\psi(x)] \, \mathrm{d}x.\end{aligned}$$

On evaluating pair differences, its sum is exactly $-\langle \bar f, (D_{B+Z} - D_B) f\rangle_{\mathrm{ch}}$. It has the four required symmetries and norm at most

$$12(1 + \rho) M^3 w_1 (w_1 + w_2) \sup_{b \text{ meeting } \square} |e^{iqZ(b)} - 1|.$$

If $q|Z| \le \frac{1}{2} q^{1-\varepsilon/2}$, its $h_0/2$ norm is at most $12 M^3 q^{1/2-\varepsilon/2}$.

Proof work. Use real accretivity and the γ_5 identity for real invertibility. Apply Schur algebra and Sylvester's determinant identity with outgoing dimension $4n_{k+1}$. Remove ω_k by reciprocal barred/unbarred phases preserving each Berezin top coefficient. Complete the block Gaussian, factor the mean interaction and keep all determinant and box normalizations. For the current reindex by the external-neighbor involution; the two halves agree. Their coefficient before chamber summation is $1/96$.

Inputs. Exact photon split and domain preservation; block field identities; weighted Gaussian completion; scalar extraction. Exports: exact normalized density and first current. At zero the determinants have size $192 n_0$ and the identity Wick kernel is 48δ; no heavy modes are removed.

Source. PDF pp. 133–142, §§13.1–13.3,13.5, Propositions 13.1–13.4, (13.1)–(13.22), (13.26)–(13.33). TeX 12018–12767, 12867–13082.

7. L. Finite-difference localization and resummation

Interface. Overview Target.

Objects and hypotheses. Use a finite torus partitioned into M-cubes. Polymers are face-connected; their net length d_M satisfies $|X|_M \le c_X(d_M(X)+1)$, $\sum_{X\supset\square} e^{-\kappa_0 d_M(X)} \le K_0$, with $c_X = 54$. For $a = M, LM$, the photon parameter family π_a consists of closed cubes of half-side a, centered at $a\mathbb{Z}^3/\ell\mathbb{Z}^3$. Fermion parameters belong to partition M-cubes. All parameters have polydisc radius $R = M^{\alpha_0}$. Containment and incidence refer to lattice sites: cubes meet when they share a lattice site. Let $P_0(X)$ be parameter cubes contained in X, let P range over subsets of parameter cubes outside $P_0(X)$, and let $Y'(X,P)$ be the union of partition cubes meeting X or the parameter cubes in $P_0(X)\cup P$. For a parameter-analytic family define

$$T_P G(X) = \int_{[0,1]^P} \partial_P G(s,X)\,\mathrm{d}s_P, \quad s=1 \text{ on } P_0(X),\ s=0 \text{ outside } P_0(X)\cup P,$$
$$\operatorname{Loc} G(Y) = \sum_{Y'(X,P)=Y} T_P G(X).$$

Assume $R - 1 \ge \exp((2L+1)^3\kappa + \log(6c_X))$.

General analytic claim. If $\|G(s,X)\| \le B_0 e^{-\kappa' d_M(X)}$, $\kappa_0 \le \kappa' \le \kappa$, and $T_P G = 0$ for disconnected Y', then

$$\sum_Y \operatorname{Loc} G(Y) = \sum_X G(1,X), \qquad \|\operatorname{Loc} G(Y)\| \le 4c_X K_0 B_0 e^{-(\kappa'-\kappa_0-1)d_M(Y)}.$$

The disconnected cancellation is a hypothesis to verify, not a consequence of this norm bound.

Proof work and use. Repeated one-variable fundamental theorems of calculus give the finite sum identity. Cauchy's formula bounds each differentiated parameter by $(R-1)^{-1}$; the geometric estimate

$$d_M(Y'(X,P)) - d_M(X) \le (2L+1)^3|P|$$

controls the new polymer weight. Count the parameter cubes and use the rooted polymer sum in M20. The resulting numerical sum has the stated factor $4c_X K_0$. This lemma assumes disconnected cancellation. In the two chamber applications, S13 proves that hypothesis from the actual component restrictions and row supports.

Source. QEM1 Section 14.1, `lem:rgnl-resummation`, `eq:rgnl-localization-sum` and `eq:rgnl-resummation-bound`; TeX 13273–13406. This is the general analytic clause of QEO1/S13, separated from its model-specific verification.

8. S13. Finite localization before the connected expansion

Model-specific. Overview Target.

Proof dependencies. P, RNO1, S07, L, and S12.

Objects and hypotheses. The incoming activity is in D's precisely defined class $\mathcal{A}_k^{x,\mathbb{C}}$, or its real subspace, with all the generator transformations and domain definitions stated in D. Use the finite-difference operation and support union of L, with all its geometric and parameter-radius hypotheses.

RG objects and hypotheses. The analytic photon and fermion families have exponential kernel/short-Hölder bounds $C_w e^{-\gamma_w d}$, local polymer decay $e^{-(\kappa+\kappa_0+3)d_M(X)}$, and are exactly restricted to active components when parameters outside are zero. A fermion H_k row at (c,g) vanishes unless every cube meeting $\{|c'-c|_\infty \le 2L\}$ is active. Local Dirac terms are $M_k(X)$. The reconstruction rows have coefficient sums at most C_T and create only external boundary pair legs. Let $e = e_k$, $h = (e^{-1/4}, e^{-1/4+\varepsilon})$, $w_W = 1$ at positive levels and $\sqrt{48}$ at zero. Input data obey $|m| \le 4e^{3/4-8\varepsilon}$, $\|E\|_k \le 4e^{1/4-7\varepsilon}$, with $E = 0$ at zero. The background slots have the local-real-gauge and complex thresholds $e^{-3/4+i\varepsilon}$, $i = 1,2,3$, the global margin $e^{\varepsilon}\theta_i$, and external thresholds $e^{-1+2\varepsilon}, e^{-1+3\varepsilon}$. For ζ in its coordinate box set

$$Z(s') = H_k^{\mathrm{ph}}(s')C_{\mathrm{ph}}C_k^{1/2}(s')\zeta, \quad |Z|_* = \max(\|Z\|, \|\partial Z\|, \|\delta_\alpha \partial Z\|) \le \tfrac{1}{2}e^{-\varepsilon/2},$$
$$B = A + C, \quad J^{\#} = H_k^{\#}(s'', B+Z) - H_k^{\#}(s'', B), \quad W^{\#} = H_k^{\#}(s'', B)(\Gamma_k(s'', B)W, \bar{W}),$$
$$r_1 = \tfrac{1}{2}e^{-3/4+9\varepsilon/2}, \quad r_2 = e^{-3/4+7\varepsilon}, \quad r_4 = e^{-1/4+3\varepsilon}.$$

Here $\psi^{\#} = (\psi, \chi; \bar{\psi}, \bar{\chi})$ lists the ordinary, pair, barred ordinary and barred pair fields in order, and $H_k^{\#}$ acts on an independent pair $(\Phi, \bar{\Phi})$. The ordinary components of $J^{\#}$ and $W^{\#}$ are denoted by $(J, \bar{J})$ and $(W^c, \bar{W}^c)$, respectively. The asymmetric fluctuation map is

$$\widetilde{\Gamma}_k(s'', B)(W, \bar{W}) = (\Gamma_k(s'', B)W, \bar{W}),$$

so $W^{\#} = H_k^{\#}\widetilde{\Gamma}_k(W, \bar{W})$. Assume $2 \le r_2, r_4 \le r_1$ and all backgrounds on these contours stay inside the stated open domains. Use $0 < 1-\alpha < \varepsilon \le 1/200$, $L \ge 4^8$, $M > 4L$ and $\kappa \ge 7\kappa_0 + 7$. The fixed positive constants are C_w, γ_w in QEO1/S05–QEO1/S06's explicit kernel estimates,

$$c_C = 3L^{9/2}, \qquad c_{\mathrm{fl}} = 61 c_X K_0, \qquad C_T = 2\exp(3/2)[1 + b_-^{-1}(54L\exp(6) + 12\exp(1) + 1)].$$

Choose $c_\gamma \ge 1$ bounding the two exponential sums over unit bonds and photon coordinate supports,

$$\sup_b \sum_{b'} \exp[-\gamma_w d(b, b')] \le c_\gamma, \qquad \sup_{\Upsilon} \sum_{\Upsilon'} \exp[-\gamma_w d(\Upsilon, \Upsilon')] \le c_\gamma.$$

They depend only on the fixed construction parameters, not on the level, side, allowed mass or charge. The stated domain assumption is the explicit QEO1/S05 conclusion $A + tZ \in \widetilde{R}_k$ and $A + C + tZ \in \widetilde{G}_k$ for $|t| \le r_1$, with the domains defined in D and QEO1/S06. It uses D's three displayed total-connection inequalities.

Claim for the finite RG expressions. Write $\mathcal{C}_r f = (2\pi i)^{-1}\int_{|t|=r} f(t)/[t(t-1)]\,\mathrm{d}t$, $m' = m + m(E)$, $E^r = RE$. Define

$$\begin{aligned}
G_1(X) &= \mathcal{C}_{r_1}\langle\bar\Psi, M_k(X, B+tZ)\Psi\rangle,\\
G_2(X) &= \mathcal{C}_{r_2}E^r(X; A+tZ, C, \psi^\# + tJ^\#\Psi),\\
G_3(\square) &= \mathcal{C}_{r_2}(-m')\langle\bar\psi + t\bar J\bar\Psi, \psi + tJ\Psi\rangle_\square,\\
E_{\le 3}(Y) &= \mathrm{sub}_{Y,B}\sum_{i\le 3}\mathrm{Loc}\,G_i(Y),\\
G^W_{\le 3}(X) &= E_{\le 3}(X; \psi^\# + W^\#),\\
G_4(X) &= \mathcal{C}_{r_4}E^r(X; A, C, \psi^\# + tW^\#),\\
G_5(\square) &= \mathcal{C}_{r_4}(-m')\langle\bar\psi + t\bar W^c, \psi + tW^c\rangle_\square,\\
F(Y) &= \mathrm{Loc}(G^W_{\le 3} + G_4 + G_5)(Y).
\end{aligned}$$

Arguments suppressed in the last four lines are unchanged. At zero G_1 is the explicit chamber current of S12, $G_2 = G_3 = G_4 = 0$, and no auxiliary Ψ or reconstruction occurs. The reconstruction $\mathrm{sub}_{Y,B}$ replaces Ψ by the bounded interior/boundary row. Both localizations have disconnected cancellation and exact sum identities. Their coefficients read only bonds meeting Y and photon coordinates supported there; F has fluctuation indices only in Y. The remaining legs have the required ordinary and admissible pair support. Moreover

$$\begin{aligned}
\|E_{\le 3}(Y)\|_{h/2} &\le 8c_X K_0 e^{1/4-5\varepsilon}\exp[-(\kappa-\kappa_0-1)d_M(Y)],\\
\|F(Y)\|_{h/4;w_W} &\le H_{\mathrm{cl}}\exp[-(\kappa-2\kappa_0-2)d_M(Y)],\\
H_{\mathrm{cl}} &= 36c_X^2K_0^2e^{1/4-5\varepsilon}.
\end{aligned}$$

F is a fixed term plus a complex-linear function of (m, E). The quantitative admission conditions used here are

$$c_{\mathrm{fl}}e^{\varepsilon},\ 3Me^{\varepsilon},\ 2c_C C_w^2 c_\gamma^2 e^{\varepsilon/4},\ 16c_X c_\gamma C_w C_T^2 M^3 e^{\varepsilon/2},\ 96M^3e^{1/4},$$
$$32C_wC_Te^{3\varepsilon/2},\ 88M^3e^{1/4-6\varepsilon},\ 2e^{5\varepsilon/2}\ \le 1,\qquad 112C_w^2c_\gamma e^{2\varepsilon}\le 3.$$

Proof work. Prove finite locality before using any connected expansion. At indicator parameter families, follow each composed row through its active component; the stronger H-row vanishing controls pair endpoints just outside X. A disconnected component contains a parameter absent from the expression, whose two finite-difference terms cancel. Apply L only after this cancellation has been verified. For numerical bounds use

$$|H^\#|_{1,\infty}\le 4C_w,\quad |J^\#|_{1,\infty}\le 16C_we^{3/4-9\varepsilon/2},\quad w_W|H^\#\widetilde\Gamma|_{1,\infty}\le 28C_w^2c_\gamma.$$

Give auxiliary fields weight C_Th_1, then pass to $h/4$ in the second localization. Apply the general bound twice.

Inputs. Weakened primitive bounds with component restriction and row vanishing; domain margins; weighted substitutions; current and reconstruction; polymer geometry. Exports: a finite local fluctuation activity and its explicit norm bound. The weakened operators are not asserted to have finite range.

Source. PDF pp. 142–149, §§14.1–14.3, Proposition 14.1, Lemma 14.2, Propositions 14.3–14.4, (14.1)–(14.23). TeX 13083–13788.

9. S14. A quantitative connected-polymer logarithm

Interface. Overview Target.

Proof dependencies. M20, S07

Objects and hypotheses. Let $\mathcal{P}$ be the collection of nonempty face-connected M-polymers. Every X_i and Y_i below belongs to $\mathcal{P}$. Use represented-net length d_M, size bound $|X|_M \le c_X(d_M(X)+1)$, and rooted counting bounds, with $c_X > 0$, $K_0 > 0$ and $\kappa_0 \ge c_X$,

$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \le K_0, \qquad \sum_{X \subset Y} e^{-\kappa_0 d_M(X)} \le K_0 |Y|_M.$$

Two polymers are incompatible when their union is face-connected, including equality. In a finite Grassmann algebra let $V(X)$ be even, with old-generator weights w and fluctuation weight $w_W = \max \nu^{-1/2}$. It has fluctuation modes only in X and depends on real box coordinates only through coordinates supported in X. The photon law is a product probability law. Compatible polymers have disjoint fluctuation and coordinate supports. Suppose

$$\|V(X)\|_{w;w_W} \le H e^{-a_2 d_M(X)}, \quad a_2 = \kappa - 2\kappa_0 - 2, \quad \kappa \ge 7\kappa_0 + 7.$$

Set $a_5 = \kappa - 3\kappa_0 - 2$, $a_6 = \kappa - 5\kappa_0 - 3$, $H_1 = 7 c_X K_0 \exp(3a_5) H$, and require

$$H \le 1, \quad 4 H K_0 c_X \exp(3a_5) \le 1, \quad z_S = 28 H_1 \exp(\kappa_0) K_0 \exp(3a_6) \le \tfrac{1}{2}.$$

Claim. For any cube union Y, define

$$K(Y) = \sum_{\substack{\{X_i\} \text{ distinct} \\ \bigcup_i X_i = Y}} \prod_i (e^{V(X_i)} - 1), \quad \mathcal{H}(Y) = \int \mathrm{d}\mu_I(W)\, \mathrm{d}\widehat{\mu}(\zeta)\, K(Y).$$

Set $K(\varnothing) = 1$, hence $\mathcal{H}(\varnothing) = 1$. Let $u^T(Y_1, \dots, Y_n)$ be the alternating sum over connected spanning subgraphs of the incompatibility graph, with $u^T(Y_1) = 1$. For each nonempty connected $Y \in \mathcal{P}$, define

$$E^{\#}(Y) = \sum_{n \ge 1} \frac{1}{n!} \sum_{\substack{Y_1, \dots, Y_n \in \mathcal{P} \\ \bigcup_i Y_i = Y}} u^T(Y_1, \dots, Y_n) \prod_i \mathcal{H}(Y_i).$$

This series is absolutely convergent in the old norm and

$$\int \mathrm{d}\mu_I\, \mathrm{d}\widehat{\mu} \exp\{\sum_{X \in \mathcal{P}} V(X)\} = \exp\{\sum_{Y \in \mathcal{P}} E^{\#}(Y)\},$$
$$\|E^{\#}(Y)\|_w \le 14 c_X^2 K_0^2 \exp(3\kappa) H e^{-(\kappa - 5\kappa_0 - 4) d_M(Y)}.$$

The norm sum of terms with $n > N$ is at most 2^{-N} times this bound. Replace each $e^V - 1$ by $\sum_{1 \le j \le N} V^j / j!$ and truncate the connected series at $n = N$. These polynomial approximants converge uniformly over all admitted families at each fixed torus side. Their finite-product error has factorial denominator $(N+1)!$ and constants depending on the finite number of polymers; no volume-uniform approximation error is asserted.

Proof work. Prove the formal connected-graph identity by component decomposition. Prove the tree bound by assigning each connected graph its least-labelled parent tree at preceding root-distance levels, and summing signs on the resulting graph intervals. Joining nets along incompatible pairs costs less than $3M$ per tree edge. Bound $K(Y)$ by $H_1 e^{-(\kappa - 3\kappa_0 - 3) d_M(Y)}$. Weighted Berezin integration and photon probability integration cost no norm; compatible supports factor. Sum a rooted tree from leaves to root using polymer counting, the factorial bound on child counts and Cayley's degree count. The degree count is the Prüfer formula $(n-2)! / \prod_i (d_i - 1)!$ for $n \ge 2$, $d_i \ge 1$, $\sum_i d_i = 2n - 2$; the $n = 1$ term is separate. Its source is Postnikov's MIT 18.212 course, notes by A. Lin, Lecture 23, Proposition 4, p. 4. The remaining geometric series has ratio $z_S \le 1/2$. Absolute convergence

justifies substitution of even Banach-algebra elements into the formal identity. Prove the polynomial approximation with its stated fixed-volume dependence.
Inputs. Weighted Berezin/Wick theorem and polymer geometry only. Exports: a local connected logarithm, its decay, geometric tail and polynomial approximants. For the chamber application take $V = F$, $H = 36c_X^2 K_0^2 e_k^{1/4-5\varepsilon}$ and $w_W = \sqrt{48}$ at zero.
Source. PDF pp. 149–152, §14.4, Proposition 14.5, (14.24)–(14.34) and its polynomial-approximant conclusion. TeX 13789–14107.

Appendix A. Detailed proof tasks and supporting statements

These common definitions specify the fields, domains, coefficient norm and symmetry conditions used throughout the target and proof tasks.

D. The activity space and its field domains

Setup. Overview Target.

Shared activity space. The nonlinear construction uses the same activity algebra as RNO1/D. We recall its complete norm, domains, support and generator actions here.

Chamber labels and evaluation. A chamber site is $x = (c, g)$, where $c \in \eta\mathbb{Z}^3/\ell\mathbb{Z}^3$, with $\eta = L^{-k}$ for a nonnegative integer k, and g ranges over the 48 signed permutation matrices in dimension three. There are four spin components and separate barred and unbarred generators. The three internal neighbours replace g by gs_i, where s_0 flips the third coordinate, s_1 exchanges the last two, and s_2 exchanges the first two. The external neighbour is $x^{(3)} = (c + \eta g e_1, g r_{e_1})$, where r_{e_1} flips the first coordinate. This neighbour convention fixes the external pairs used below. For a connection B, $U_B(x, x')$ is the exponential of ie_k times the arithmetic mean of its six coordinate-order path sums, each with the mesh factor η. It is one within a photon cluster. The pair generators remain independent until the evaluation

$$\chi(x, x') = \frac{U_{A+C}(x, x')\psi(x') - \psi(x)}{\max(|c - c'|_\infty, \eta)^\alpha}$$

and its barred version with inverse transport. Statements about an evaluated activity refer to this substitution.

The common setting. Define the activity space $\mathcal{A}_k^{x,\mathbb{C}}$, with real subspace $\mathcal{A}_k^x$. Fix odd $L \geq 5$, $\eta = L^{-k}$, $M = L^m \geq L^2$, $\ell/(LM)$ odd and at least 13, $0 < \varepsilon \leq 1/200$, $0 < \alpha < 1$, and $0 < e_k < 1$. An M-polymer is small if $d_M(X) \leq L$, with an injective covering lift. Put $\mathrm{Vol}(X) = M^3|X|_M$. Balanced means that each monomial has equally many barred and unbarred generators. An activity $E(X; A, C)$ is a balanced polynomial in independent ordinary and pair Grassmann legs, with holomorphic coefficients reading only bonds meeting X. Ordinary legs are over X; an allowed pair has all six coordinate-order paths in X, or is an external nearest-neighbor pair with an endpoint over X. Its separation is $d_X = \max(|c - c'|_\infty, \eta)$, and the pair range is $0 < d_X \leq M$; intracluster pairs are included. Index measures are $\eta^3/48$ and $(\eta^3/48)^2$, while generator weights are $h_1 = e_k^{-1/4}$, $h_2 = e_k^{-1/4+\varepsilon}$. Ordinary kernels use the antiperiodic convention in each chamber argument; pair indices are simultaneous period classes of their two endpoints. The coefficient norm of the separately antisymmetric kernels is

$$\|E(X; A, C)\|_{h_k} = \sum_{n,m} \frac{h_1^n h_2^m}{n!m!} \int |E_{nm}(X; A, C; \xi; \zeta)| \, \mathrm{d}\xi \, \mathrm{d}\zeta.$$

To specify its domains, put

$$\begin{aligned} \partial_\nu B_\mu(c) &= \eta^{-1}[B_\mu(c + \eta e_\nu) - B_\mu(c)], \\ \|B\|_1 &= \sup |B_\mu(c)|, \quad \|B\|_2 = \sup |\partial_\nu B_\mu(c)|, \\ \|B\|_3 &= \sup_{\mu,\nu} \sup_{0<d(c,c')\leq 1} \frac{|\partial_\nu B_\mu(c') - \partial_\nu B_\mu(c)|}{d(c, c')^\alpha}, \qquad \theta_i = e_k^{-3/4+i\varepsilon}. \end{aligned}$$

The distance is periodic sup distance; a local norm uses only its contained bonds and difference cells. $R_k = \{B : \|B\|_i < \theta_i,\ i = 1, 2, 3\}$. The lattice gradient is $d\lambda(c, c') = \eta^{-1}(\lambda(c') - \lambda(c))$. A field belongs to $\widetilde{R}_k$ if it is $A_0 + A_1$, A_0 real, $A_1 \in R_k$, and on every enlarged cube of side $(2L + 1)M$, $A_0 = A_\square + d\lambda_\square$ with real $A_\square, \lambda_\square$ and $\|A_\square\|_{i,\square\dagger} < \theta_i$. The smaller domain $\widetilde{R}_k^-$ retains the same local

real representatives and replaces the global bounds on A_1 by $\|A_1\|_i < (1-e_k^{\varepsilon})\theta_i$. The external domain is

$$\mathfrak{X}_k = \{C : \|C\|_1 < e_k^{-1+2\varepsilon},\ \|C\|_2 < e_k^{-1+3\varepsilon}\}.$$

The activity norm is precisely

$$\|E\|_k = \sup_X \exp(\kappa d_M(X)) \sup_{(A,C)\in \widetilde{R}_k\times\mathfrak{X}_k} \|E(X;A,C)\|_{h_k}.$$

Assume the polymer counting bound $\sum_{X\supset\square} e^{-\kappa_0 d_M(X)} \le K_0$, $\kappa \ge \kappa_0$.

Here is the complete symmetry package. Use $\gamma_\mu = \sigma_3\otimes\sigma_\mu$, $\gamma_5 = \sigma_1\otimes 1_2$, $\mathsf{C} = i\gamma_2$, and $\varkappa = \gamma_5\mathsf{C}$. For a spin lift S generated by the four γ-matrices and a translation $a\in M\mathbb{Z}^3$, let $\widetilde{g}(c,g) = (r(S)c+a, r(S)g)$, where $S\gamma_\mu S^{-1} = r(S)_{\mu\mu}\gamma_\mu$. Define

$$a_{\widetilde{g}}\psi(x) = S\psi(\widetilde{g}^{-1}x), \qquad a_{\widetilde{g}}\bar{\psi}(x) = (S^{-1})^T\bar{\psi}(\widetilde{g}^{-1}x).$$

For pair generators transform both endpoints and use the same spin matrices. G_λ multiplies an unbarred ordinary or pair leg by $\exp(ie_k\lambda(c))$, with c its first photon point, and a barred leg by its inverse. Define on ordinary and pair legs

$$\mathrm{Cg}\,\psi = \mathsf{C}\bar{\psi}, \quad \mathrm{Cg}\,\bar{\psi} = -\mathsf{C}\psi, \qquad \mathbf{K}\psi = \varkappa\psi, \quad \mathbf{K}\bar{\psi} = \varkappa\bar{\psi},$$

with $\mathbf{K}$ antilinear. The complex activity class requires

$$\begin{aligned} a_{\widetilde{g}}E(\widetilde{g}X;\widetilde{g}A,\widetilde{g}C) &= E(X;A,C),\\ \mathrm{Cg}\,E(X;-A,-C) &= E(X;A,C),\\ G_{\lambda+\lambda'}E(X;A^1,C^1) &= E(X;A,C) \end{aligned}$$

whenever the two pairs are in the domain and $A^1 = A - d\lambda$, $C^1 = C - d\lambda'$ on bonds at distance at most one from X, for real λ,λ'. Its real subspace additionally requires

$$\mathbf{K}E(X;-\overline{A},-\overline{C}) = E(X;A,C).$$

All equations act on independent generators before evaluation. For the reconstruction application in RNO1, the explicit conditions

$$e_k^{\varepsilon} \le \tfrac{1}{4}, \qquad e_k^{1/4} + e_k^{\varepsilon} \le 1, \qquad 4e_k^{1/4+2\varepsilon} + 2e_k^{3\varepsilon} \le 1$$

give $\widetilde{R}_k + \mathfrak{X}_k \subset \widetilde{G}_k$, where $\widetilde{G}_k$ has a global complex bound $e_k^{-1+\varepsilon}$ and local real-gauge sup bounds $e_k^{-1+\varepsilon}$ on every $11M$ cube. They also give $e_k\|\operatorname{Im}(A+C)\|_\infty \le 1$ and $e_k\|d(A+C)\|_\infty \le 1$.

Source. QEM1 Sections 12.1–12.2, TeX 10142–10370; the margin-reduced domain is used at 13164–13262. The activity notation $\mathcal{A}_k^{x,\mathbb{C}}$ here denotes the space $K_k^{x,\mathbb{C}}$ of QEM1. This setup is shared with RNO1/D.